

How Devices Shape Mental Effort in Digital Document Reading: An Eye-Tracking Study

Kumushini Thennakoon, Yasasi Abeysinghe, Pasindu Thenahandi, Lawrence Obiuevwi
Vikas Ashok, Sampath Jayarathna

Department of Computer Science, Old Dominion University, Norfolk, VA, USA
kumushini@cs.odu.edu, yasasi@cs.odu.edu, pasindu@cs.odu.edu, lobi001@odu.edu
vganjigu@cs.odu.edu, sampath@cs.odu.edu

Abstract—The shift from print to digital reading environments has introduced new challenges for understanding how device interfaces influence mental effort during reading. While prior work has often relied on controlled user studies, this study investigates device-dependent mental effort using the Reading in the Wild dataset collected with Project Aria. We analyze eye tracking data during reading tasks on four commonly used digital devices: laptops, tablets, mobile phones, and wall-mounted screens, using fixation duration as the key eye-tracking measure to understand mental effort. Our results indicate that device type significantly shapes mental effort: wall-mounted screens elicited the longest fixations, laptops showed intermediate values, and handheld devices exhibited the lowest and most comparable levels of effort. These findings highlight how ergonomic differences such as viewing distance and display size influence visual processing and mental efforts during digital reading. Our work advances the understanding of device-specific mental effort and offers insights for designing digital reading environments that emphasize usability.

Index Terms—Digital documents, Mental effort, Eye-tracking

I. INTRODUCTION

The increasing prevalence of digital reading environments has transformed how individuals interact with written information [1]. Digital documents have become the primary medium for knowledge acquisition in academic, professional, and everyday contexts. Unlike traditional print, digital media allow users to access content via different device interfaces like laptops, mobile phones, tablets, and desktop screens. Understanding the visual behavior and mental workload of readers is crucial for improving engaged reading experiences and for designing device-optimized interfaces.

Cognitive load plays an essential role in the reading comprehension process by keeping relevant information accessible in short-term memory and integrating it with information retrieved from long-term memory [2]. When we interact with digital media to read, our brains will navigate not only the content but also the technological interface, creating additional layers of processing that can impact mental effort [3], [4]. Processing pixelated text and eye strain from exposure to the screen can affect the reader's ability to maintain focus and comprehension [5]. This effort can vary according to the interface or the device we use. Understanding these variations is essential for the user to have more effective digital reading experiences and to recognize when cognitive fatigue could im-

pact comprehension and learning outcomes in our increasingly digital world.

Recent advances in physiological sensing, such as eye-tracking and biometric monitoring, offer promising avenues to understand mental effort during reading tasks [6], [7]. Eye movement patterns (e.g., fixations, saccades, pupil dilation) can reveal processing difficulty and attention distribution [8], [9] while engaged in reading activities. Among these eye-movement metrics fixation duration is one of the most important eye-movement indicators of cognitive load [10], [11].

In this paper, we examine the relationship between reading digital documents and mental effort using fixation duration data. We check if there is a difference in the mental effort when reading digital documents with 4 device interfaces. Using egocentric data [12] collected in natural environments adds ecological validity to the study and additionally provides new insights into how different digital platforms influence mental effort during reading.

II. RELATED WORK

Technological advancements have a significant impact on the reading practices of digital document environments. Karman and Aylin [13] reported that reading from screens can be slower, more tiring, and more demanding compared to printed documents, due to factors like resolution and display characteristics. Furthermore, the visual quality of documents can vary across devices, as mobile phones, tablets, and PCs differ in screen size, resolution, and pixel density. These differences influence text clarity, image rendering, and overall visual comfort, thereby affecting the reader's cognitive effort and comprehension [14]. Furthermore, reading behaviors and mental effort have been shown to differ across domains and levels of researcher expertise, reinforcing the role of document structure and content in navigation strategies [15]. Collectively, these studies highlight the complexity of digital reading as a socio-technical practice influenced by medium, interface design, user expertise, and social context.

Mental effort, a central construct in cognitive load research, can be assessed using different methods, each offering distinct insights and trade-offs. These metrics are commonly classified into three categories: (1) Subjective Measures, which rely on participants' self-reports of perceived mental effort or difficulty, typically gathered through questionnaires, rating

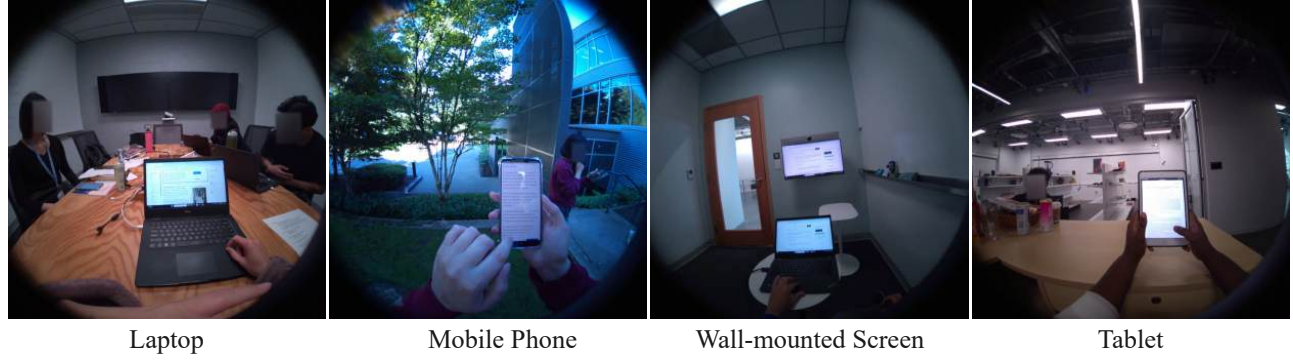


Fig. 1. Example egocentric video frames extracted from the recordings captured using Project Aria glasses. The frames illustrate participants reading documents on four device types.

scales, or Likert-type assessments such as NASA-TLX [16]; (2) Behavioral Measures, which use objective indicators of task performance that reflect cognitive load, such as task completion time, error rates, reading comprehension accuracy, and navigation patterns. Longer task times or increased errors typically indicate higher cognitive effort. Behavioral measures are useful because they are directly related to task outcomes and can be collected unobtrusively. For example, Dillon et al. [14] used subjective effort ratings alongside task performance to measure cognitive load while reading digital documents of varying image quality; and (3) Physiological Measures, which capture cognitive load through biological signals. Ayres et al. [17] examined the validity and sensitivity of physiological techniques, identifying measures across four categories: heart and lungs, eyes, skin, and brain. Among these, eye-tracking measures have consistently proven to be highly sensitive, followed by heart rate, skin responses, and brain activity.

Eye-tracking offers measures for quantifying mental effort. Consistent with earlier claims that digital reading demands greater effort, Delgado and Salmerón [18] provided eye-tracking evidence showing that readers exhibit higher cognitive effort when processing digital texts compared to print. Metrics such as pupil diameter, blink rates, fixations, saccades, and ocular pressure are particularly sensitive indicators of cognitive load [17]. Applications extend to assistive contexts, where pupil metric and blink-based metrics have been used to predict fatigue in eye-typing users [19], and positioned pupillometry was proposed as a translational biomarker for effort-related brain states [20]. Collectively, these works demonstrate that eye-tracking metrics are reliable and non-invasive measures for evaluating cognitive load, including during digital reading tasks.

III. METHODOLOGY

A. Dataset

We used the *Reading in the Wild* dataset [21], published by Meta, which contains more than 100 hours of reading and non-reading videos in diverse and realistic scenarios. The dataset was collected using Project Aria glasses [22] and includes

three modalities: egocentric video recording, eye gaze data, and head pose data. This dataset also provides the raw data in a VRS file [23] format, containing videos from the world-view camera, eye-tracking cameras, and mono-scene cameras. For this study, we utilized the *engaged reading* subset, in which participants were asked to read a document such as a research paper or a blog post—while wearing Project Aria glasses. Before the reading tasks, each participant performed an in-session eye gaze calibration using the built-in eye tracking calibration of the Aria glasses. This enables the glasses to capture personalized eye gaze data with better accuracy of eye gaze tracking by compensating for user-specific biases [22]. Figure 1 shows example video frames from recordings during the data collection step. Participants were reading digital documents using devices, including mobile phones, tablets, laptops, and wall-mounted screens, providing multimodal egocentric data in naturalistic environments. The sitting positions of the participants and the device handling behaviors were slightly different for each device. Table I records the specifics in device handling. The reading materials include blog posts, e-books, news articles, research papers, and Wikipedia etc. Participants were 25-45 years old and included both male and female readers.

TABLE I
OVERVIEW OF HOW PARTICIPANTS HELD ELECTRONIC DEVICES AND POSITIONED THEMSELVES WHILE READING

Device	Specifics in device handling
Laptop	Reader sits near the device (≈ 30 cm)
Tablet	Reader holds the device (≈ 30 cm)
Mobile phone	Reader holds the device (≈ 30 cm)
Screen	Reader sits 3m-4m away from the device

B. Processing Eye-tracking Data

Gaze data were extracted through the machine perception service (MPS) [22] and provided in yaw angle and pitch angle in radians. We used personalized gaze data derived from the in-session calibration data collected during recording to obtain more accurate individual gaze estimations. Using the

MPS personalized gaze data and VRS file, we converted the angular gaze data into x and y coordinates to represent 2D gaze positions using Project Aria tools [23]. Each recording included data from both the gaze calibration step and the reading task. The calibration data were excluded using the calibration timestamp parameter.

C. Calculating Fixation Duration

To identify fixations from 2D gaze data, we applied a modified version of the I-DT (Identification by Dispersion-Threshold) algorithm [24]. In our case, the input consisted of gaze coordinates projected onto an RGB image along with corresponding timestamps. In this modification, we defined the temporal window of 100 ms and the spatial threshold of 50 pixels. Then we clustered all the gaze points within both windows and identified the fixation. The 50-pixel threshold was chosen because the visual interface (device used for reading) occupied a relatively small area within the 1408×1408 pixel image, ensuring localized but not overly restrictive fixations. For each fixation, start time, end time, duration, and centroid coordinates were recorded.

D. Analyzing fixation duration

After calculating the fixations, we concatenated the data from all 4 devices into a single dataset. To reduce bias, we randomly sampled 1,000 fixation duration records per device. A one-way ANOVA was applied to examine differences in mean fixation durations across the four devices with equal group sizes ($n=1000$), ensuring sufficient statistical power for the analysis.

IV. RESULTS

A one-way ANOVA was conducted to examine the effect of device type on fixation duration, $F(3, 3996) = 95.36$, $p < 0.05$, indicating that mean fixation durations differ across devices.

Fixation duration is commonly used as an indication of mental effort [25], as longer fixations indicate increased cognitive processing during reading. Figure 2 shows that reading documents from a wall-mounted screen exhibited the highest mean fixation duration among all device types, followed by laptops, while tablets showed the shortest.

TABLE II
ONE-WAY ANOVA RESULTS FOR FIXATION DURATION ACROSS DEVICES

Source	Sum of Squares	df	F	p-value
C(device)	$1.43.66 \times 10^8$	3	95.36	1.30×10^{-59}
Residual	1.99×10^9	3996	–	–

Following the ANOVA, a Tukey HSD post hoc test identified pairwise differences in fixation durations across devices. The results revealed that most device pairs show a significant difference ($p < 0.05$), except for mobile phones and tablets. Specifically, laptop vs. screen, laptop vs. mobile, laptop vs. tablet, screen vs. mobile, and screen vs. tablet all differed significantly. These results indicate that the mobile and tablet devices form a group showing a comparable mental effort

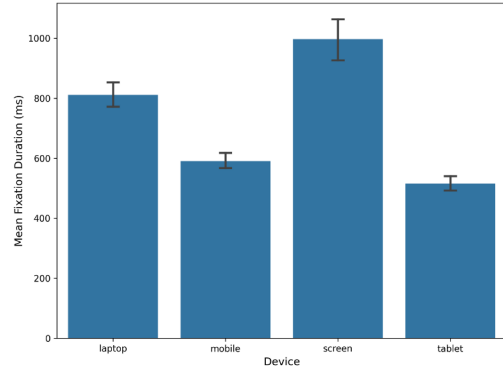


Fig. 2. Comparison of mean fixation duration for participants reading on four devices. Error bars indicate the 95% confidence interval for each device.

to read digital documents. Figure 3 presents the confidence intervals for the pairwise comparisons between devices.

TABLE III
MULTIPLE COMPARISON OF MEANS - TUKEY HSD, FWER = 0.05

Group 1	Group 2	Mean Diff	p-adj	Lower	Upper	Reject
laptop	mobile	-220.8919	0.000	-302.1923	-139.5915	True
laptop	screen	185.4312	0.000	104.1308	266.7317	True
laptop	tab	-295.5607	0.000	-376.8611	-214.2603	True
mobile	screen	406.3231	0.000	325.0227	487.6235	True
mobile	tab	-74.6688	0.085	-155.9693	6.6316	False
screen	tab	-480.9920	0.000	-562.2924	-399.6916	True

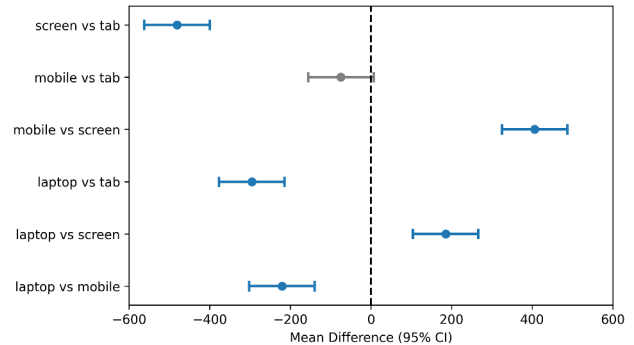


Fig. 3. Pairwise mean fixation duration differences between devices obtained using Tukey's HSD test. Error bars show 95% confidence intervals

V. DISCUSSION

Our findings demonstrate that device type significantly influences fixation duration and, by extension, mental effort during digital reading [26]. The one-way ANOVA confirmed overall differences, while Tukey HSD post hoc tests revealed significant pairwise device contrasts in fixation durations, except between mobile and tablet devices.

When examining mean fixation durations across the four device types, wall-mounted screens elicited the longest fixations, suggesting the highest mental engagement or effort, followed

by laptops. In contrast, tablets exhibited the shortest fixations, indicating relatively lower cognitive load during reading tasks.

Several contextual factors likely contributed to these observed differences. As outlined in our methodology, participants were positioned in close proximity to laptops, tablets, and mobile devices, facilitating comfortable viewing conditions. However, participants were seated 3–4 meters away from the wall-mounted screen, creating suboptimal reading conditions for detailed text content such as blog posts. This increased viewing distance likely necessitated greater visual and cognitive effort to process textual information [27]. Additionally, the wall-mounted display setup exposed participants to broader environmental distractions within their visual field compared to the focused viewing experience offered by hand-held devices.

Our analysis relied on first-person video recordings and corresponding gaze data, with video frames capturing details from the surrounding environment. Given the engaged nature of the reading task, we assumed that participants were consistently directing their attention toward the target device. However, this assumption may not always hold, particularly in more ecologically valid settings where attention might naturally drift. The most closely related eye tracking measure to the mental effort is pupil dilation, but Project Aria glasses, which are not specifically made for eye tracking research, cannot capture pupil dilation directly.

VI. CONCLUSION

This study highlights the critical importance of understanding how different devices shape cognitive processing in naturalistic reading scenarios. The observed differences in mental effort between the device types have significant practical implications. The results provide insights to designers to consider device-specific differences in cognitive load while reading, such as viewing distance and screen size, when optimizing usability and user experience. In future work, we plan to integrate automated object detection to verify gaze on the target device, incorporate additional physiological measures, and analyze task comprehension alongside eye movements.

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